R91 | Locked outcomes: delivery recovered, research gate not passed

R91A: utility CE only | R91B: utility links also for risk/R-Drop | seed2, logical batch1024

Model / split	Top1 (%)	mean_cost	SBS cost	Oracle cost	vs SBS (%)	vs Oracle (%)	actual regret (%)
R84A / VAL	71.085185	11.416611477	11.427978473	11.371691902	-0.194190	0.393443	0.403494
R84A / TEST	71.235802	11.373843044	11.385299246	11.329932369	-0.197053	0.386140	0.394847
R88B / VAL	71.058025	11.416634105	11.427978473	11.371691902	-0.192477	0.395188	0.405128
R88B / TEST	71.220988	11.373838370	11.385299246	11.329932369	-0.195592	0.387648	0.396339
R91A / VAL	71.072840	11.416850456	11.427978473	11.371691902	-0.191608	0.396065	0.405976
R91A / TEST	71.293827	11.373669575	11.385299246	11.329932369	-0.196579	0.386647	0.395084
R91B / VAL	71.091358	11.416996733	11.427978473	11.371691902	-0.190903	0.396778	0.408374
R91B / TEST	71.408642	11.374208708	11.385299246	11.329932369	-0.194204	0.389022	0.398966

Equal weight across18 problems. SBS fitted on TRAIN81000; Oracle = per-instance legal minimum; native/FP64 ind.

Joint VAL gate: Top1 strictly increases, cost and actual regret do not increase versus BOTH historical references.

Arm / reference (VAL)	Top1 delta pp	mean_cost delta	regret delta pp	Reference pass	Joint pass
A / R84A	-0.012346	+0.000238979	+0.002482	False	False
A / R88B	+0.014815	+0.000216352	+0.000849	False	False
B / R84A	+0.006173	+0.000385257	+0.004880	False	False
B / R88B	+0.033333	+0.000362629	+0.003247	False	False

Both arms FAIL the fixed joint VAL gate. B raises TEST Top1 but worsens TEST cost/regret versus A and R84A.

R84A and same-role R88B are historical references, not newly matched original-loss controls; single seed only.

All18 use frozen NEW90k splits81000/4500/4500. TEST informed prior R&D; these are not untouched blind results.

A lock: update37440 / fc889238acd827fa | B lock: update40320 / cdcb5f1da1e3c30c. Both trained57600 updates.

R91 | Frozen target36 recovery and matched B-minus-A tradeoff

R91A: utility CE only | R91B: utility links also for risk/R-Drop | seed2, logical batch1024

Arm / ref / split	Winners	Correct	Selected	Recall %	Precision %	Rescued	Lost	New false	New false regret pp	Target regret delta pp
A/R84A/VAL	4625	10	24	0.2162	41.6667	3	4	10	0.0003307	+0.0002639
A/R84A/TEST	4598	11	26	0.2392	42.3077	2	4	6	0.0000917	+0.0007054
A/R88B/VAL	4625	10	24	0.2162	41.6667	4	3	8	0.0002458	-0.0002737
A/R88B/TEST	4598	11	26	0.2392	42.3077	2	3	7	0.0001421	+0.0007082
B/R84A/VAL	4625	18	39	0.3892	46.1538	9	2	19	0.0015882	-0.0005853
B/R84A/TEST	4598	26	53	0.5655	49.0566	16	3	19	0.0010863	-0.0002167
B/R88B/VAL	4625	18	39	0.3892	46.1538	11	2	16	0.0007227	-0.0011229
B/R88B/TEST	4598	26	53	0.5655	49.0566	17	3	21	0.0012113	-0.0002140

Same36 R84A VAL-defined method/problem groups; never reselected. VAL4625 / TEST4598 true target winners.

B recovers more true winners, but TEST recall remains0.5655%; target-local regret gains do not imply global gains.

B minus A	Top1 delta pp	Corrected	Harmed	mean_cost delta	actual regret delta pp
VAL	+0.018519	992	977	+0.000146277	+0.002398
TEST	+0.114815	990	897	+0.000539133	+0.003881

Recommendation: retain R84A as reference; neither R91 arm establishes joint cost/regret plus recovery improvement.

Full reports: each arm/full_model_statistics/MODEL_STATISTICS.pdf (NSS4 + ALL/per-task SBS/Oracle metrics).

selection_distributions.pdf:18 pages, paired selection/winner bars; target_recovery_and_train.pdf: cached TRAIN/VAL/TEST.

Original font failure, workers/controllers, checkpoint locks and evaluations remain unchanged; see report_only_recovery_complete.json.